Mathematical Analysis II
Week 5 Homework (March 31)

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Section 9.2

29. 若 $u = F(x, y)$，F 任意二阶偏导存在，而 $x = r\cos\varphi, y = r\sin\varphi$。证明：

$$\left(\frac{\partial u}{\partial r}\right)^2 + \left(\frac{1}{r}\frac{\partial u}{\partial \varphi}\right)^2 = \left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2.$$

证. 由链式法则得

$$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial r} = F'_x \cos\varphi + F'_y \sin\varphi,$$

$$\frac{\partial u}{\partial \varphi} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial \varphi} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial \varphi} = -F'_x r\sin\varphi + F'_y r\cos\varphi.$$

因此

$$\frac{1}{r}\frac{\partial u}{\partial \varphi} = -F'_x \sin\varphi + F'_y \cos\varphi.$$

从而

$$\begin{aligned} LHS &= F_x'^2 \cos^2\varphi + 2F'_xF'_y \cos\varphi\sin\varphi + F_y'^2 \sin^2\varphi \\ &\quad + F_x'^2 \sin^2\varphi - 2F'_xF'_y \sin\varphi\cos\varphi + F_y'^2 \cos^2\varphi \\ &= (F_x'^2 + F_y'^2)(\cos^2\varphi + \sin^2\varphi) \\ &= F_x'^2 + F_y'^2 = RHS. \end{aligned}$$

Q.E.D.

30. 试证：方程 $\dfrac{\partial^2 u}{\partial x^2} + 2\dfrac{\partial^2 u}{\partial x \partial y} - 3\dfrac{\partial^2 u}{\partial y^2} + 2\dfrac{\partial u}{\partial x} + 6\dfrac{\partial u}{\partial y} = 0$ 经变化 $\xi = x + y, \eta = 3x - y$ 后变成 $\dfrac{\partial^2 u}{\partial \eta \partial \xi} + \dfrac{1}{2}\dfrac{\partial u}{\partial \xi} = 0$。（其中二阶偏导数均连续）

证. 计算偏导数：

$$\frac{\partial \xi}{\partial x} = 1, \quad \frac{\partial \xi}{\partial y} = 1, \quad \frac{\partial \eta}{\partial x} = 3, \quad \frac{\partial \eta}{\partial y} = -1.$$

因此

$$\frac{\partial u}{\partial x} = u_\xi + 3u_\eta, \quad \frac{\partial u}{\partial y} = u_\xi - u_\eta.$$

进而求得二阶偏导数：

$$\begin{aligned}\frac{\partial^2 u}{\partial x^2} &= u_{\xi\xi}+3u_{\xi\eta}+3u_{\eta\xi}+9u_{\eta\eta}=u_{\xi\xi}+6u_{\xi\eta}+9u_{\eta\eta},\\ \frac{\partial^2 u}{\partial x\partial y} &= u_{\xi\xi}+3u_{\xi\eta}-u_{\eta\xi}-3u_{\eta\eta}=u_{\xi\xi}+2u_{\xi\eta}-3u_{\eta\eta},\\ \frac{\partial^2 u}{\partial y^2} &= u_{\xi\xi}-u_{\xi\eta}-u_{\eta\xi}+u_{\eta\eta}=u_{\xi\xi}-2u_{\xi\eta}+u_{\eta\eta}.\end{aligned}$$

代入原方程，得

$$\begin{aligned}LHS = {} & u_{\xi\xi}+6u_{\xi\eta}+9u_{\eta\eta}+2(u_{\xi\xi}+2u_{\xi\eta}-3u_{\eta\eta})-3(u_{\xi\xi}-2u_{\xi\eta}+u_{\eta\eta})\\ &+2(u_\xi+3u_\eta)+6(u_\xi-u_\eta)\\ = {} & 16u_{\xi\eta}+8u_\xi.\end{aligned}$$

即

$$16\frac{\partial^2 u}{\partial\eta\partial\xi}+8\frac{\partial u}{\partial\xi}=0.$$

两边同时除以 16，即得证。

Q.E.D.

32. 设变换 $\begin{cases}u=x-2y,\\ v=x+ay\end{cases}$ 可把方程 $6\frac{\partial^2 z}{\partial x^2}+\frac{\partial^2 z}{\partial x\partial y}-\frac{\partial^2 z}{\partial y^2}=0$ 简化为 $\frac{\partial^2 z}{\partial u\partial v}=0$。求常数 a。（其中二阶偏导数均连续）

解. 计算偏导数：

$$\frac{\partial u}{\partial x}=1,\quad \frac{\partial u}{\partial y}=-2,\quad \frac{\partial v}{\partial x}=1,\quad \frac{\partial v}{\partial y}=a.$$

因此

$$\frac{\partial z}{\partial x}=z_u+z_v,\quad \frac{\partial z}{\partial y}=-2z_u+az_v.$$

进而求得二阶偏导数：

$$\begin{aligned}\frac{\partial^2 z}{\partial x^2} &= z_{uu}+2z_{uv}+z_{vv},\\ \frac{\partial^2 z}{\partial x\partial y} &= -2z_{uu}+(a-2)z_{uv}+az_{vv},\\ \frac{\partial^2 z}{\partial y^2} &= 4z_{uu}-4az_{uv}+a^2z_{vv}.\end{aligned}$$

代入原方程，得

$$\begin{aligned}LHS &= 6(z_{uu}+2z_{uv}+z_{vv})+(-2z_{uu}+(a-2)z_{uv}+az_{vv})-(4z_{uu}-4az_{uv}+a^2z_{vv})\\ &= (10+5a)z_{uv}+(6+a-a^2)z_{vv}.\end{aligned}$$

为使方程简化为 $\frac{\partial^2 z}{\partial u\partial v}=0$，需满足

$$\begin{cases}6+a-a^2=0,\\ 10+5a\neq 0.\end{cases}$$

解得 $a=3$。

33. 求方程 $\dfrac{\partial z}{\partial y}=x^2+2y$ 满足条件 $z(x,x^2)=1$ 的解 $z=z(x,y)$。

解. 对 y 积分得

$$z=x^2y+y^2+C(x).$$

代入边界条件 $z(x,x^2)=1$，得

$$x^2\cdot x^2+(x^2)^2+C(x)=1,$$

即

$$x^4+x^4+C(x)=1.$$

因此

$$C(x)=1-2x^4.$$

故所求解为

$$z(x,y)=x^2y+y^2-2x^4+1.$$

34. 设 $u=u(x,y)$，当 $y=x^2$ 时有 $u=1$, $\dfrac{\partial u}{\partial x}=x$，求当 $y=x^2$ 时的 $\dfrac{\partial u}{\partial y}$。

解. 构造辅助函数 $\varphi(x)=u(x,x^2)=1$，则 $\varphi'(x)=0$。

由链式法则得

$$\begin{aligned}\varphi'(x)&=\left.\frac{\partial u}{\partial x}\right|_{y=x^2}+\left.\frac{\partial u}{\partial y}\right|_{y=x^2}\cdot\frac{dy}{dx}\\&=x+\left.\frac{\partial u}{\partial y}\right|_{y=x^2}\cdot\frac{d}{dx}(x^2)\\&=x+\left.\frac{\partial u}{\partial y}\right|_{y=x^2}\cdot 2x=0.\end{aligned}$$

当 $x\neq 0$ 时，两边除以 $2x$，解得

$$\left.\frac{\partial u}{\partial y}\right|_{y=x^2}=-\frac{1}{2}.$$

35. 设 $u=u(x,y)$ 满足方程 $\dfrac{\partial^2 u}{\partial x^2}-\dfrac{\partial^2 u}{\partial y^2}=0$ 以及条件 $u(x,2x)=x$, $u'_x(x,2x)=x^2$，求 $u''_{xx}(x,2x)$, $u''_{xy}(x,2x)$, $u''_{yy}(x,2x)$。（其中二阶偏导数均连续）

解. 由方程知

$$u_{xx}=u_{yy}.$$

构造辅助函数 $\varphi(x)=u(x,2x)=x$，则

$$\varphi'(x)=u_x+2u_y=x^2+2u_y=1,$$

解得

$$u_y=\frac{1-x^2}{2}.$$

进而

$$\varphi''(x) = u_{xx} + 2u_{xy} + 2u_{yx} + 4u_{yy} = 5u_{xx} + 4u_{xy} = 0.$$

又令 $\psi(x) = u_x(x, 2x) = x^2$，则

$$\psi'(x) = u_{xx} + 2u_{xy} = 2x.$$

联立方程组

$$\begin{cases} 5u_{xx} + 4u_{xy} = 0, \\ u_{xx} + 2u_{xy} = 2x, \end{cases}$$

解得

$$\begin{cases} u_{xx} = -\dfrac{4}{3}x, \\ u_{xy} = \dfrac{5}{3}x. \end{cases}$$

因此

$$u''_{xx}(x, 2x) = -\frac{4}{3}x, \quad u''_{xy}(x, 2x) = \frac{5}{3}x, \quad u''_{yy}(x, 2x) = u''_{xx}(x, 2x) = -\frac{4}{3}x.$$

36. 求下列复合函数的微分 $\mathrm{d}u$

(2) $u = f(\xi, \eta)$, $\xi = xy$, $\eta = \dfrac{x}{y}$;

解. 计算中间变量的微分：

$$\mathrm{d}\xi = y\mathrm{d}x + x\mathrm{d}y, \quad \mathrm{d}\eta = \frac{1}{y}\mathrm{d}x - \frac{x}{y^2}\mathrm{d}y.$$

因此

$$\begin{aligned} \mathrm{d}u &= \frac{\partial u}{\partial \xi}\mathrm{d}\xi + \frac{\partial u}{\partial \eta}\mathrm{d}\eta \\ &= f_\xi(y\mathrm{d}x + x\mathrm{d}y) + f_\eta\left(\frac{1}{y}\mathrm{d}x - \frac{x}{y^2}\mathrm{d}y\right) \\ &= \left(f_\xi y + \frac{f_\eta}{y}\right)\mathrm{d}x + \left(f_\xi x - \frac{f_\eta x}{y^2}\right)\mathrm{d}y \\ &= \left(f_\xi y + \frac{f_\eta}{y}\right)\mathrm{d}x + \left(f_\xi - \frac{f_\eta}{y^2}\right)x\mathrm{d}y. \end{aligned}$$

(4) $u = f(x, \xi, \eta)$, $\xi = x^2 + y^2$, $\eta = x^2 + y^2 + z^2$;

解. 计算中间变量的微分：

$$\mathrm{d}\xi = 2x\mathrm{d}x + 2y\mathrm{d}y, \quad \mathrm{d}\eta = 2x\mathrm{d}x + 2y\mathrm{d}y + 2z\mathrm{d}z.$$

因此

$$\begin{aligned} \mathrm{d}u &= f_x\mathrm{d}x + f_\xi\mathrm{d}\xi + f_\eta\mathrm{d}\eta \\ &= f_x\mathrm{d}x + f_\xi(2x\mathrm{d}x + 2y\mathrm{d}y) + f_\eta(2x\mathrm{d}x + 2y\mathrm{d}y + 2z\mathrm{d}z) \\ &= (f_x + 2xf_\xi + 2xf_\eta)\mathrm{d}x + 2(f_\xi + f_\eta)y\mathrm{d}y + 2zf_\eta\mathrm{d}z. \end{aligned}$$

37. 求在球坐标下的 Laplace 方程的形式。

解. 在直角坐标系下，*Laplace* 方程为

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0.$$

在球坐标系下，坐标变换关系为

$$\begin{cases} r = \sqrt{x^2 + y^2 + z^2}, \\ \theta = \arctan \dfrac{\sqrt{x^2 + y^2}}{z}, \\ \varphi = \arctan \dfrac{y}{x}. \end{cases}$$

由链式法则，

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial u}{\partial \theta}\frac{\partial \theta}{\partial x} + \frac{\partial u}{\partial \varphi}\frac{\partial \varphi}{\partial x} = \frac{\partial u}{\partial r}\frac{x}{r} + \frac{\partial u}{\partial \theta}\frac{xz}{r^2\sqrt{x^2 + y^2}} - \frac{\partial u}{\partial \varphi}\frac{y}{x^2 + y^2}.$$

同理可得

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r}\frac{y}{r} + \frac{\partial u}{\partial \theta}\frac{yz}{r^2\sqrt{x^2 + y^2}} + \frac{\partial u}{\partial \varphi}\frac{x}{x^2 + y^2},$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial r}\frac{z}{r} - \frac{\partial u}{\partial \theta}\frac{\sqrt{x^2 + y^2}}{r^2}.$$

由于

$$\begin{cases} x = r \sin\theta \cos\varphi, \\ y = r \sin\theta \sin\varphi, \\ z = r \cos\theta, \end{cases}$$

代入上述表达式，得

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}\sin\theta\cos\varphi + \frac{\partial u}{\partial \theta}\frac{\cos\theta\cos\varphi}{r} - \frac{\partial u}{\partial \varphi}\frac{\sin\varphi}{r\sin\theta},$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r}\sin\theta\sin\varphi + \frac{\partial u}{\partial \theta}\frac{\cos\theta\sin\varphi}{r} + \frac{\partial u}{\partial \varphi}\frac{\cos\varphi}{r\sin\theta},$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial r}\cos\theta - \frac{\partial u}{\partial \theta}\frac{\sin\theta}{r}.$$

写成矩阵形式，即

$$\begin{pmatrix} \dfrac{\partial u}{\partial x} \\ \dfrac{\partial u}{\partial y} \\ \dfrac{\partial u}{\partial z} \end{pmatrix} = \underset{\underset{A}{\downarrow}}{\begin{pmatrix} \sin\theta\cos\varphi & \dfrac{\cos\theta\cos\varphi}{r} & -\dfrac{\sin\varphi}{r\sin\theta} \\ \sin\theta\sin\varphi & \dfrac{\cos\theta\sin\varphi}{r} & \dfrac{\cos\varphi}{r\sin\theta} \\ \cos\theta & -\dfrac{\sin\theta}{r} & 0 \end{pmatrix}} \begin{pmatrix} \dfrac{\partial u}{\partial r} \\ \dfrac{\partial u}{\partial \theta} \\ \dfrac{\partial u}{\partial \varphi} \end{pmatrix}.$$

记

$$\nabla_{xyz}=\begin{pmatrix}\dfrac{\partial}{\partial x}\\ \dfrac{\partial}{\partial y}\\ \dfrac{\partial}{\partial z}\end{pmatrix},\quad \nabla_{r\theta\varphi}=\begin{pmatrix}\dfrac{\partial}{\partial r}\\ \dfrac{\partial}{\partial \theta}\\ \dfrac{\partial}{\partial \varphi}\end{pmatrix}.$$

因此 $\nabla_{xyz}u=A\nabla_{r\theta\varphi}u$。

由 $\nabla^2u=\nabla_{xyz}^T\nabla_{xyz}u$，得

$$\nabla^2u=\nabla_{r\theta\varphi}^TA^TA\nabla_{r\theta\varphi}u+\sum_{1\le i,j\le 3}(\partial_iA_{ij})\partial_j.$$

经过计算，得到

$$A^TA=\begin{pmatrix}1&0&0\\0&\dfrac{1}{r^2}&0\\0&0&\dfrac{1}{r^2\sin^2\theta}\end{pmatrix}.$$

记 $B=\displaystyle\sum_{i=1}^{3}\sum_{j=1}^{3}(\partial_iA_{ij})\partial_j$，则

$$B(u)=\frac{2}{r}\frac{\partial u}{\partial r}.$$

因此

$$\begin{aligned}\nabla^2u&=\nabla_{r\theta\varphi}^TA^TA\nabla_{r\theta\varphi}u+B(u)\\&=\begin{pmatrix}\dfrac{\partial}{\partial r}&\dfrac{\partial}{\partial \theta}&\dfrac{\partial}{\partial \varphi}\end{pmatrix}\begin{pmatrix}1&0&0\\0&\dfrac{1}{r^2}&0\\0&0&\dfrac{1}{r^2\sin^2\theta}\end{pmatrix}\begin{pmatrix}\dfrac{\partial u}{\partial r}\\ \dfrac{\partial u}{\partial \theta}\\ \dfrac{\partial u}{\partial \varphi}\end{pmatrix}+\frac{2}{r}\frac{\partial u}{\partial r}\\&=\frac{\partial^2u}{\partial r^2}+\frac{1}{r^2}\frac{\partial^2u}{\partial\theta^2}+\frac{1}{r^2\sin^2\theta}\frac{\partial^2u}{\partial\varphi^2}+\frac{2}{r}\frac{\partial u}{\partial r}.\end{aligned}$$

由于

$$\frac{\partial^2u}{\partial r^2}+\frac{2}{r}\frac{\partial u}{\partial r}=\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial u}{\partial r}\right),$$

因此球面坐标系下的 *Laplace* 方程的形式为

$$\nabla^2u=\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial u}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2u}{\partial\theta^2}+\frac{1}{r^2\sin^2\theta}\frac{\partial^2u}{\partial\varphi^2}=0.$$

Section 9.3

1. 证明下列方程在指定点的附近对 y 有唯一解，并求出 y 对 x 在该点处的一阶和二阶导数

(1) $x^2+xy+y^2=7$，在 $(2,1)$ 处.

证. 令 $F(x,y)=x^2+xy+y^2-7\in C^1$，则 $F(2,1)=4+2+1-7=0$。

计算偏导数得 $F_y=x+2y$，因此 $F_y(2,1)=4\neq 0$。由隐函数存在定理知，在 $(2,1)$ 附近对 y 有唯一解。

Q.E.D.

解. 由隐函数求导公式得

$$\frac{\mathrm{d}y}{\mathrm{d}x}=-\frac{F_x}{F_y}=-\frac{2x+y}{x+2y}.$$

因此在点 $(2,1)$ 处，

$$y'=\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{(2,1)}=-\frac{5}{4}.$$

对 $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ 关于 x 求导，得

$$y''=-\frac{(2+y')(x+2y)-(2x+y)(1+2y')}{(x+2y)^2}.$$

代入 $(2,1)$ 及 $y'=-\dfrac{5}{4}$，计算得

$$y''|_{(2,1)}=-\frac{\frac{3}{4}\cdot 4+5\cdot\frac{3}{2}}{4^2}=-\frac{21}{32}.$$

(2) $x\cos xy=0$，在 $(1,\frac{\pi}{2})$ 处.

证. 令 $F(x,y)=x\cos xy\in C^1$，则 $F(1,\frac{\pi}{2})=\cos\frac{\pi}{2}=0$。

计算偏导数得 $F_y=-x^2\sin xy$，因此 $F_y(1,\frac{\pi}{2})=-1\neq 0$。由隐函数存在定理知，在 $(1,\frac{\pi}{2})$ 附近对 y 有唯一解。

Q.E.D.

解. 由隐函数求导公式得

$$\frac{\mathrm{d}y}{\mathrm{d}x}=-\frac{F_x}{F_y}=\frac{\cos xy-xy\sin xy}{x^2\sin xy}.$$

因此在点 $(1,\frac{\pi}{2})$ 处，

$$y'=\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{(1,\frac{\pi}{2})}=-\frac{\pi}{2}.$$

对 $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ 关于 x 求导，经过化简得

$$y''=\frac{2y'}{x}+\frac{-\cos xy(2x\sin xy+x^2(y+xy')\cos xy)}{x^4\sin^2 xy}.$$

代入 $(1,\frac{\pi}{2})$ 及 $y'=-\dfrac{\pi}{2}$，计算得

$$y''|_{(1,\frac{\pi}{2})}=-2y'=\pi.$$

2. 求由下列方程所确定的隐函数的导数.

(1) $\sin(xy)-\mathrm{e}^{xy}-x^2y=0$，求 $\dfrac{\mathrm{d}y}{\mathrm{d}x}$;

解. 方程两边关于 x 求导，得

$$y\cos xy + xy'\cos xy - y\mathrm{e}^{xy} - xy'\mathrm{e}^{xy} - 2xy - x^2y' = 0.$$

整理得

$$xy'(\cos xy - \mathrm{e}^{xy} - x) = -y\cos xy + y\mathrm{e}^{xy} + 2xy.$$

因此

$$\frac{\mathrm{d}y}{\mathrm{d}x} = y' = \frac{-y\cos xy + y\mathrm{e}^{xy} + 2xy}{x\cos xy - x\mathrm{e}^{xy} - x^2}.$$

(3) $x^y = y^x$，求 $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ 和 $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}$;

解. 两边取对数得 $y\ln x = x\ln y$。

两边关于 x 求导，得

$$y'\ln x + \frac{y}{x} = \ln y + \frac{xy'}{y}.$$

整理得

$$y'(y\ln x - x) = x\ln y - y.$$

因此

$$\frac{\mathrm{d}y}{\mathrm{d}x} = y' = \frac{x\ln y - y}{y\ln x - x}.$$

进一步求二阶导数，得

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{(1-y')(y'\ln x + \frac{y}{x})}{y\ln x - x}.$$

(5) $\dfrac{x}{z} = \ln\dfrac{z}{y}$，求 $\dfrac{\partial z}{\partial x}, \dfrac{\partial z}{\partial y}$;

解. 令 $F(x,y,z) = \dfrac{x}{z} + \ln y - \ln z = 0$。

计算偏导数得

$$F_x = \frac{1}{z}, \quad F_y = \frac{1}{y}, \quad F_z = -\frac{x}{z^2} - \frac{1}{z}.$$

由隐函数求导公式得

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = \frac{\frac{1}{z}}{\frac{x}{z^2} + \frac{1}{z}} = \frac{z}{x+z},$$

$$\frac{\partial z}{\partial y} = -\frac{F_y}{F_z} = \frac{\frac{1}{y}}{\frac{x}{z^2} + \frac{1}{z}} = \frac{z^2}{(x+z)y}.$$

(7) $F(xz, yz) = 0$，求 $\dfrac{\partial z}{\partial x}, \dfrac{\partial z}{\partial y}$.

解. 令 $u = xz$, $v = yz$。

方程两边关于 x 求偏导，得

$$F_u\left(z + x\frac{\partial z}{\partial x}\right) + F_v\left(y\frac{\partial z}{\partial x}\right) = 0.$$

解得

$$\frac{\partial z}{\partial x} = -\frac{F_u z}{F_u x + F_v y}.$$

方程两边关于 y 求偏导，得

$$F_u\left(x\frac{\partial z}{\partial y}\right) + F_v\left(z + y\frac{\partial z}{\partial y}\right) = 0.$$

解得

$$\frac{\partial z}{\partial y} = -\frac{F_v z}{F_u x + F_v y}.$$

第九章综合习题

2. 设 $f(x,y,z) = F(u,v,w)$，其中 $x^2 = vw$, $y^2 = wu$, $z^2 = uv$。求证：

$$x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} + z\frac{\partial f}{\partial z} = u\frac{\partial F}{\partial u} + v\frac{\partial F}{\partial v} + w\frac{\partial F}{\partial w}.$$

证. 由 $x^2 = vw$, $y^2 = wu$, $z^2 = uv$ 得 $yz = u\sqrt{vw} = ux$，因此

$$u = \frac{yz}{x}.$$

同理可得

$$v = \frac{xz}{y}, \quad w = \frac{xy}{z}.$$

由链式法则，

$$\frac{\partial f}{\partial x} = \frac{\partial F}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial F}{\partial v}\frac{\partial v}{\partial x} + \frac{\partial F}{\partial w}\frac{\partial w}{\partial x} = -\frac{yz}{x^2}F_u + \frac{z}{y}F_v + \frac{y}{z}F_w.$$

因此

$$x\frac{\partial f}{\partial x} = -\frac{yz}{x}F_u + \frac{xz}{y}F_v + \frac{xy}{z}F_w = -uF_u + vF_v + wF_w.$$

同理可得

$$y\frac{\partial f}{\partial y} = uF_u - vF_v + wF_w, \quad z\frac{\partial f}{\partial z} = uF_u + vF_v - wF_w.$$

将上述三式相加，得

$$LHS = 2uF_u - uF_u + 2vF_v - vF_v + 2wF_w - wF_w = uF_u + vF_v + wF_w = RHS.$$

Q.E.D.

3. 若函数 $u = f(x,y,z)$ 满足恒等式 $f(tx,ty,tz) = t^k f(x,y,z)$ $(t > 0)$，则称 $f(x,y,z)$ 为 k 次齐次函数。试证下述关于齐次函数的欧拉定理：可微函数 $f(x,y,z)$ 为 k 次齐次函数的充要条件是：

$$xf'_x(x,y,z) + yf'_y(x,y,z) + zf'_z(x,y,z) = kf(x,y,z).$$

证.

- 必要性

  因为 $f(tx,ty,tz)=t^k f(x,y,z)$，两边关于 t 求导，得

  $$xf'_x(tx,ty,tz)+yf'_y(tx,ty,tz)+zf'_z(tx,ty,tz)=kt^{k-1}f(x,y,z).$$

  令 $t=1$，得

  $$xf'_x(x,y,z)+yf'_y(x,y,z)+zf'_z(x,y,z)=kf(x,y,z).$$

  必要性得证。

- 充分性

  构造辅助函数 $\varphi(t)=\dfrac{f(tx,ty,tz)}{t^k}$。

  对 t 求导，得

  $$\varphi'(t)=\frac{(xf'_x(tx,ty,tz)+yf'_y(tx,ty,tz)+zf'_z(tx,ty,tz))t^k-kf(tx,ty,tz)t^{k-1}}{t^{2k}}.$$

  代入条件 $xf'_x(x,y,z)+yf'_y(x,y,z)+zf'_z(x,y,z)=kf(x,y,z)$，得

  $$\varphi'(t)=\frac{kf(tx,ty,tz)t^k\cdot\frac{1}{t}-kf(tx,ty,tz)t^{k-1}}{t^{2k}}=0.$$

  因此 $\varphi(t)$ 恒为常数，故

  $$\varphi(t)=\frac{f(tx,ty,tz)}{t^k}=\varphi(1)=f(x,y,z),$$

  即 $f(tx,ty,tz)=t^k f(x,y,z)$。充分性得证。

Q.E.D.